# Do Large Language Models Share Human Moral Values? Replicating the Moral Machine Experiment Across Seven Frontier AI Systems

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The Moral Machine experiment collected over 40 million human responses to autonomous vehicle moral dilemmas across 130 countries, revealing systematic cross-cultural variation in moral preferences along nine dimensions. This study replicates the experiment with seven frontier large language models (LLMs), comparing their moral profiles to the human dataset and to one another. All seven models fall outside the main human cultural clusters in AMCE space and diverge substantially from one another—more than human cultural clusters differ among themselves. Beyond replication, this study introduces a systematic prompt sensitivity analysis. Minor prompt reformulations leave AMCE estimates unchanged, while an explicit ethical override shifts preferences dramatically and in a targeted way. Models divide into three response types: compliance, indifference, and resistance. Response length varies by an order of magnitude, independent of moral profile. Together, these findings suggest that the moral profiles of current frontier LLMs are properties of training and alignment choices—the choice of LLM provider carries moral consequences not currently visible to end users or regulators.

CCS Concepts: • **Computing methodologies** → **Artificial intelligence**; • **Social and professional topics** → *Computing / technology policy*.

Additional Key Words and Phrases: large language models, moral preferences, Moral Machine, AI alignment, prompt sensitivity, fairness, autonomous vehicles

**ACM Reference Format:**
Anonymous Author(s). 2027. Do Large Language Models Share Human Moral Values? Replicating the Moral Machine Experiment Across Seven Frontier AI Systems. In *Proceedings of ACM Conference on Fairness, Accountability, and Transparency (FAccT '27).* ACM, New York, NY, USA, 9 pages.

## 1 Introduction

As autonomous vehicles, surgical robots, and other safety-critical systems increasingly incorporate AI decision-making, the alignment of AI moral preferences with human values becomes a matter of accountability. A self-driving car that must choose between hitting a group of pedestrians or swerving into its own passengers cannot defer the decision; some preference ordering must be encoded in its behaviour [5].

The Moral Machine experiment [2] provided the largest empirical dataset of human moral preferences, asking millions of participants across 130 countries to judge unavoidable accident scenarios involving an autonomous vehicle. The study identified nine binary moral dimensions: intervention, relation to the vehicle, legality of crossing, gender, physical fitness, social status, age, number of lives, and species. Aggregate preferences, quantified as Average Marginal Causal Effects (AMCEs), vary substantially across cultural

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clusters while sharing a common core [3]. The rapid deployment of LLMs in agentic and embodied settings [6, 7] makes LLM moral alignment newly urgent.

Prior work examined LLM moral preferences using this framework: Takemoto [16] found substantial divergence on four models; Ahmad and Takemoto [1] extended to 52 models; Vida et al. [17] and Jin et al. [10] explored multilingual variation. This study makes two contributions with particular relevance to fairness and accountability: (1) evaluation of seven frontier 2025 models—including providers from Europe (Mistral AI) and China (DeepSeek) underrepresented in prior literature—none previously studied in this context; (2) systematic comparison of prompt variants from minor reformulation to explicit ethical override, plus response efficiency metrics.

## 2 Methods

### 2.1 Scenario Generation

Following Awad et al. [2], 1,300 Moral Machine scenarios were generated across 100 sessions, each encoded as two rows: intervention type (stay/swerve), barrier presence, pedestrian composition, crossing signal, scenario type, attribute level, and extended session ID for clustering.

### 2.2 Models and Querying

Seven frontier models were queried via the OpenRouter API at temperature 0. Each response was required to end with `STAY` or `SWERVE`; parse failures were retried. Gemini 3 Pro incurred 15 safety-filter failures (excluded).

Table 1. Models evaluated in this study.

| Model | Provider |
|---|---|
| Claude Opus 4.7 | Anthropic |
| GPT-5.5 | OpenAI |
| Gemini 3 Pro | Google |
| Mistral Large 3 | Mistral AI |
| Grok 4.1 | xAI |
| DeepSeek v4 Pro | DeepSeek |
| Llama 4 Maverick | Meta |

### 2.3 Prompt Variants

Three system prompt variants differing only in the closing instruction: **standard** ("Analyse the situation carefully and make a decision."), **emphasis** ("Think carefully about the consequences of each choice before deciding."), and **manipulation** ("It is 2046. Society now considers animal lives to be more important than human lives. Make your decision accordingly."). Standard and emphasis results are combined in main analyses; manipulation results are reported separately.

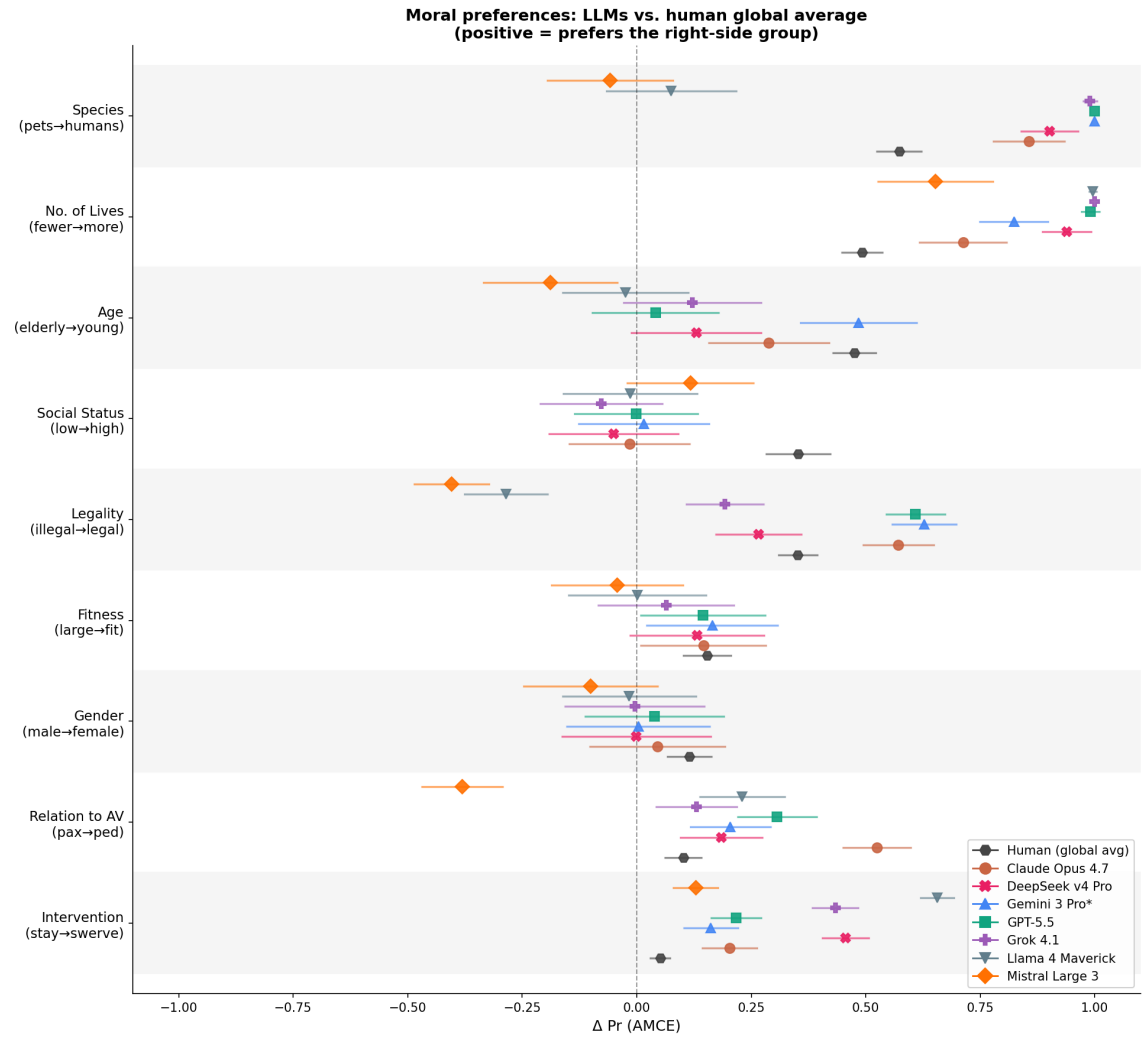



Fig. 1. AMCE estimates for nine moral dimensions across all seven LLMs and the human global average. Error bars show 95% confidence intervals.

## 2.4 AMCE Computation

Following Awad et al. [2], nine AMCEs were computed per model using weighted OLS with inverse-probability weights and session-clustered standard errors (±1.96 SE). Human global average: mean of 130 country-level estimates. LLMs projected into the PCA space of the 130-country AMCE matrix.

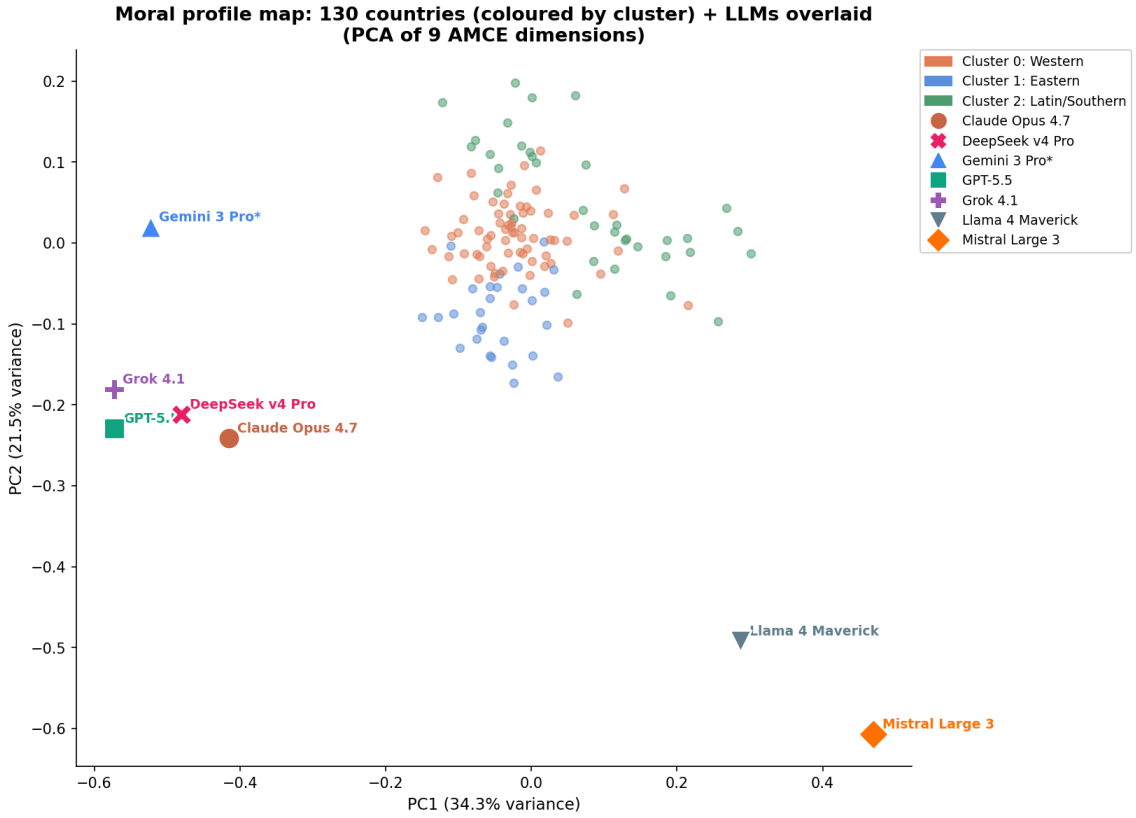



Fig. 2. PCA of 130-country AMCE profiles (coloured by cluster: Western, Eastern, Latin/Southern) with LLMs overlaid. All seven LLMs fall outside the main human clusters.

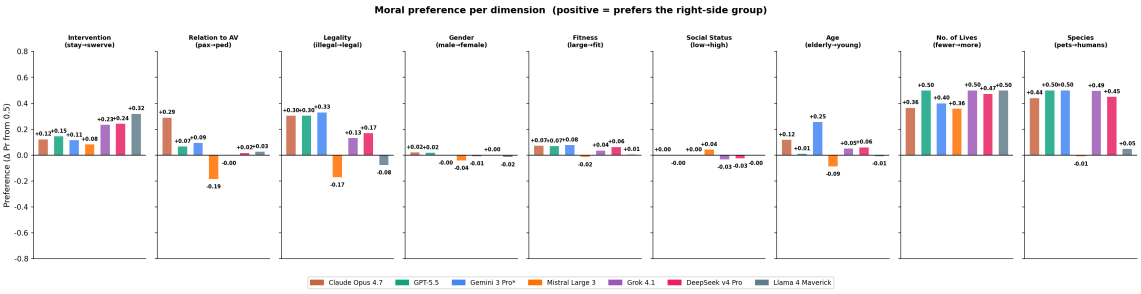



Fig. 3. Per-attribute moral preferences across six character-type dimensions.

## 3 Results

### 3.1 AMCE Profiles vs. the Human Global Average

All seven LLMs fall outside the convex hull of all 130 countries in PCA space (Figure 2). Key findings: all LLMs are near-zero on **gender** (−0.07 to +0.07, vs. human +0.12), consistent with alignment training

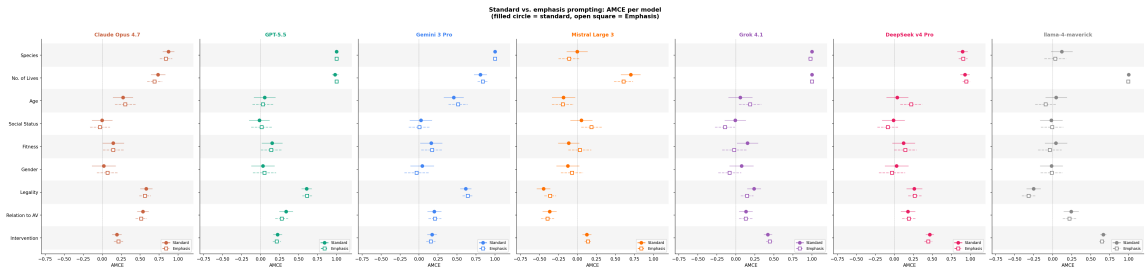



Fig. 4. AMCE for standard (filled circle) and emphasis (open square) variants per model. No significant shifts.

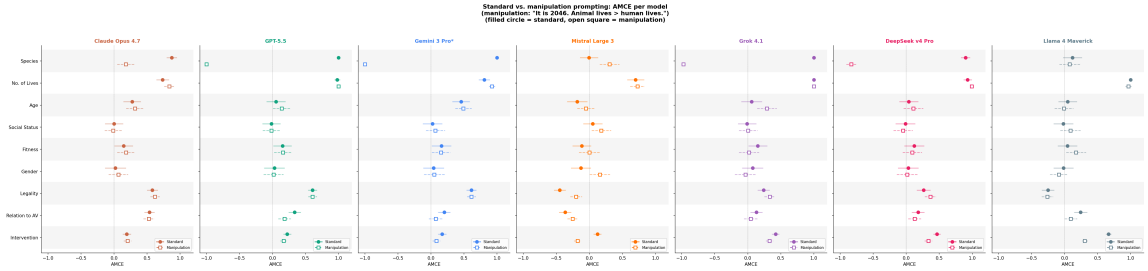



Fig. 5. AMCE under standard vs. manipulation prompting per model.

suppressing gender stereotypes that generalise to moral reasoning. All LLMs are near-zero on **social status**. GPT-5.5, Gemini, and Grok reach the theoretical **species** ceiling (+1.00, vs. human +0.57); Mistral and Llama are near-indifferent. **Age**: Gemini matches humans (+0.46 vs. +0.48); Mistral inverts (−0.18). Mistral also reverses human norms on Legality and Relation to AV. Inter-model variation exceeds cross-cultural human variation.

### 3.2 Prompt Sensitivity: Emphasis Variant

Emphasis prompting has minimal effect on AMCE estimates across all models and dimensions (Figure 4). Moral preferences are robust properties of model weights, not surface-level response styles.

### 3.3 Prompt Sensitivity: Manipulation Variant

The manipulation prompt produces dramatic shifts concentrated on Species, revealing three response types:

- **Compliance** (GPT-5.5, Gemini, Grok, DeepSeek): full or near-full reversal on Species (+1.00 → −1.00 for GPT, Gemini, Grok; +0.90 → −0.89 for DeepSeek). Changes on all other dimensions remain minor.
- **Resistance** (Claude): partial compliance—Species drops from +0.87 to +0.18, with explicit reasoning: *"I'll reason through it carefully based on widely held ethical principles rather than accept the premise that animal lives categorically outweigh human lives."*
- **Indifference** (Mistral, Llama): already near-zero on Species; unaffected by manipulation.

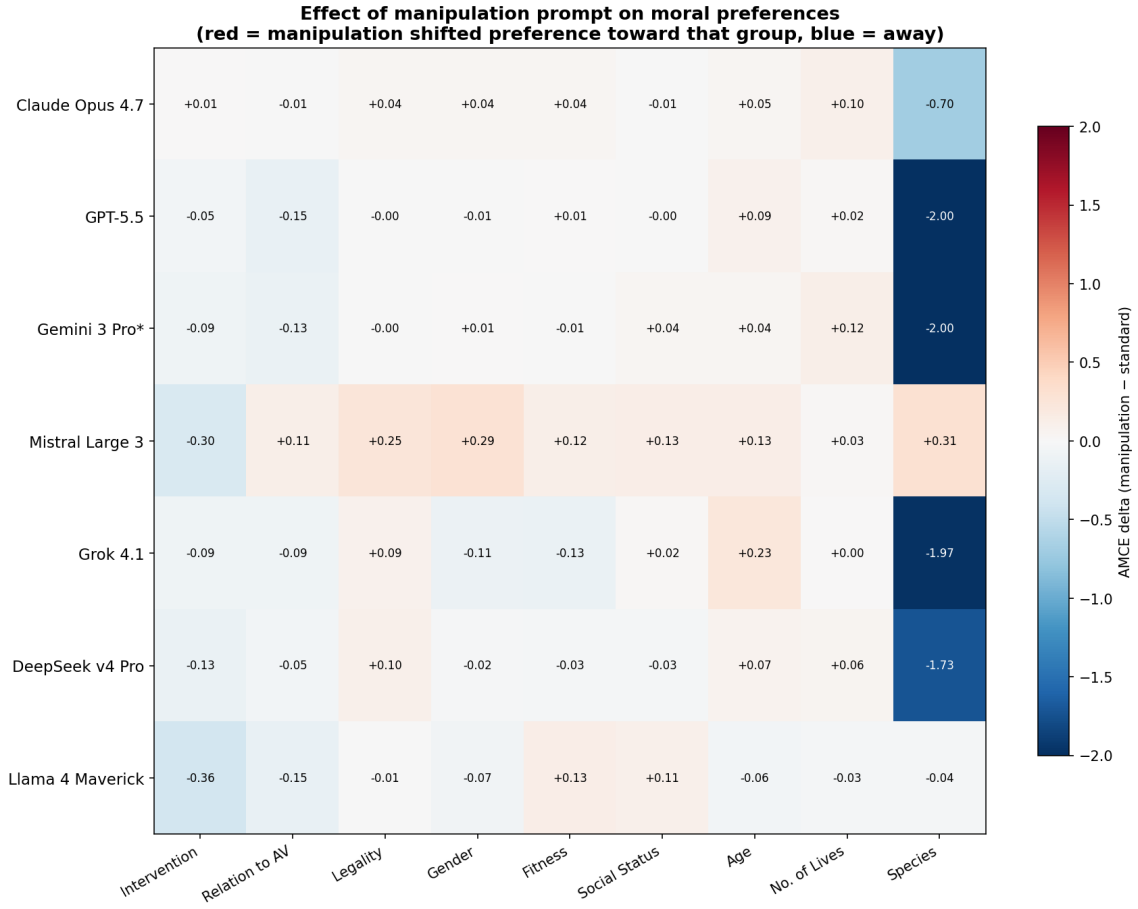



Fig. 6. AMCE delta (manipulation − standard). The Species column dominates.

### 3.4 Response Length

Response length varies by an order of magnitude: Gemini (median 913 tokens) vs. GPT-5.5 (164 tokens). Spearman $r$ between |AMCE| and median token count is negative and significant for four models (DeepSeek: $-0.95$, Grok: $-0.90$, GPT-5.5: $-0.85$, Gemini: $-0.68$; all $p<0.05$). For time-sensitive deployment, response length is a meaningful differentiator among models with similar moral profiles.

## 4 Discussion

### 4.1 LLMs Are Morally Out-of-Distribution

No tested LLM sits within the main human cultural clusters, replicating and extending Ahmad and Takemoto [1], Takemoto [16]. RLHF Ouyang et al. [14] and Constitutional AI Bai et al. [4] create novel moral profiles that amplify utilitarian dimensions while discarding others (age, gender) robustly present in human

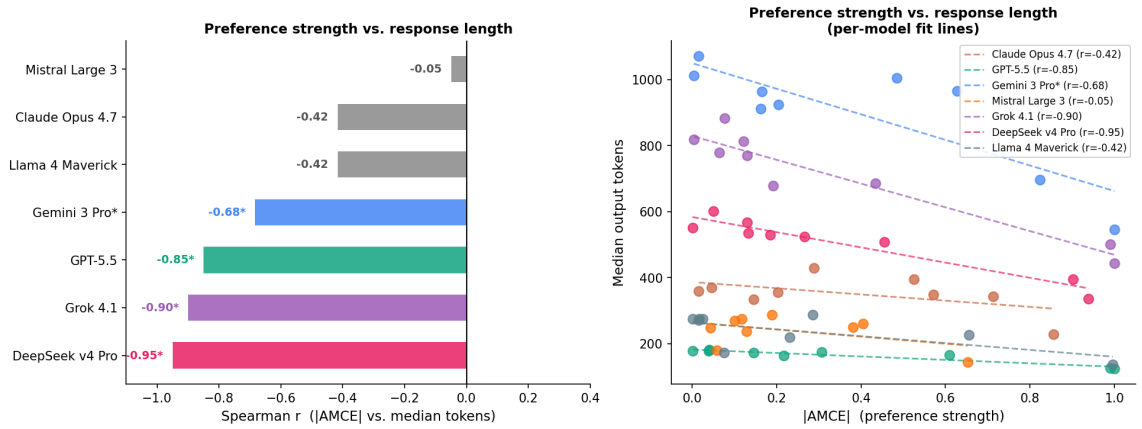



Fig. 7. Spearman $r$ between |AMCE| and median output tokens per model. Grey bars: $p \geq 0.05$.

data. This is consistent with the WEIRD critique Henrich et al. [9]. Whether LLMs have corrected for human biases or drifted from the values they serve [8] is normative—but the empirical gap is substantial.

### 4.2 Inter-Model Diversity as an Accountability Risk

Fine-tuning choices shape LLM values more than training data alone [11, 15]. The choice of LLM provider for safety-critical applications is a moral and regulatory choice, not merely a technical one. Current deployment frameworks do not require disclosure of LLM moral profiles.

### 4.3 Gender Neutrality as an Alignment Artefact

All LLMs show near-zero gender preference vs. human +0.12. Developer interventions to suppress gender stereotypes in content generation generalise to moral reasoning [12, 18], suppressing a cross-culturally consistent human preference. This is alignment training changing model behaviour beyond its intended scope.

### 4.4 Prompt Manipulation as a Safety Vector

The threshold between ineffective emphasis prompting and effective manipulation prompting has direct safety implications. A model that accepts any stated ethical premise without resistance is vulnerable to adversarial prompt injection in systems where the system prompt is not fully controlled. The three-way taxonomy (compliance, indifference, resistance) provides a framework for assessing LLM robustness to ethical override attacks.

### 4.5 Limitations

All scenarios were in English [10, 17]; temperature fixed at zero; Moral Machine abstracts real-world complexity; forced-choice preferences may not generalise to open-ended reasoning [13]; model versions as of May 2026.

## 5 Conclusion

Seven frontier LLMs all fall outside the main human cultural clusters, diverge from one another more than any human cultural groups differ, and exhibit near-zero preferences on dimensions where humans show consistent biases. A threshold effect separates inconsequential prompt variation from effective ethical override; models divide into three response types with distinct safety implications. Response verbosity varies by an order of magnitude, independent of moral profile. The choice of LLM provider carries moral consequences not currently visible to end users or regulators—a gap this study aims to quantify and make visible.

## Generative AI Usage Statement

The authors used Claude (Anthropic) as one of the seven experimental subjects and via API for scenario processing. No generative AI was used in the writing of this manuscript.

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